OPERATING SYSTEMS PRACTICALS

Easy-to-Understand Assignment

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Assignment Topics	System Information, Linux Bootable USB, macOS Settings, Ubuntu Installation, Windows 10 Installation Guide
Purpose	Learn basic operating-system setup and configuration

This assignment explains five operating-system practical tasks in simple language. It includes instructions, safety notes, expected results, and places to record observations. Perform the tasks on a device you own or have permission to use. Some tasks can be completed in a virtual machine if you do not have spare hardware.

Introduction

An operating system (OS) is the main software that helps a computer work. Windows, Linux, and macOS are examples. The OS manages the processor, memory, storage, files, applications, and connected devices. Learning how to view system information and install or configure an OS is useful for computer maintenance and IT support.

Before you start

• Back up important files before changing partitions or installing an operating system.
• Use official download websites for operating-system ISO files and installation tools.
• A clean installation can erase files. Read each installer screen carefully before confirming any disk changes.
• Use a spare USB drive for a bootable installer; creating one normally erases the USB drive's contents.
• If you only need to practise, a virtual machine is often safer than changing your real computer.

Practical 1: Find OS Version, Processor, and RAM in Windows

Aim

To open System Information and capture a screenshot showing the Windows version, processor, and installed RAM.

What the details mean

OS version: identifies the Windows edition, version, and build. **Processor:** shows the CPU model, which performs calculations and runs instructions. **Installed RAM:** shows the computer's main working memory, used by programs while they run.

Steps

1. Sign in to your Windows laptop or PC.
2. Press **Windows + R** to open the Run box.
3. Type **msinfo32** and press Enter.
4. In System Information, select **System Summary**.
5. Find **OS Name**, **Version**, **Processor**, and **Installed Physical Memory (RAM)**. The labels may vary slightly.
6. Make the window large enough to show the required details. If needed, widen the Item and Value columns.
7. Press **Windows + Shift + S**, select the part of the screen containing those details, and save the screenshot. Alternatively, press **Alt + PrtScn** to capture the active window, then paste it into Paint and save it.
8. Name the image something clear, such as **System_Information_Dhwanit.png**.

Expected result

A screenshot clearly shows the Windows OS details, processor model, and installed RAM. Use your own computer's actual information; do not guess or copy another person's specifications.

Practical 2: Download Linux and Create a Bootable USB

Aim

To obtain an official Linux ISO file and prepare a USB installer using Rufus or BalenaEtcher. If a spare USB is unavailable, the ISO can be tested in a virtual machine.

Important terms

Linux distribution: a complete Linux operating system, such as Ubuntu or Fedora. **ISO file:** a disc-image file containing the installation media. **Bootable USB:** a USB drive prepared so a computer can start from it and launch the installer.

Steps to download the ISO

1. Visit the official Ubuntu website at ubuntu.com/download/desktop or Fedora at fedoraproject.org/workstation/download.
2. Choose the desktop version suitable for your computer. For most recent Intel or AMD PCs, the standard 64-bit PC image is appropriate.

3. Download the ISO file and wait until the download finishes. Keep it in a folder you can find easily, such as Downloads.
4. Check that the download completed and that the filename ends in **.iso**.

Option A: Make a bootable USB with Rufus

1. Back up the USB drive. Creating the installer will erase the drive's existing contents.
2. Download Rufus from its official website: rufus.ie.
3. Insert the USB drive and open Rufus.
4. Under Device, carefully select the correct USB drive. Check its size so you do not accidentally select another disk.
5. Choose **SELECT** and open the downloaded Linux ISO.
6. Leave the recommended settings unless the Linux distribution's instructions say otherwise, then choose **START**.
7. Read any warning about erasing data and continue only if you selected the correct USB. Wait for the status to show that the process is complete, then safely eject the USB.

Option B: Use BalenaEtcher

Download Etcher from etcher.balena.io. Open it, select the Linux ISO, select the correct USB drive, and choose Flash. Confirm the destination carefully. Wait for flashing and verification to finish, then safely eject the drive.

Option C: Use a virtual machine instead

If you do not have a spare USB, install VirtualBox from virtualbox.org or use another trusted virtual-machine tool. Create a new virtual machine, choose Linux/Ubuntu as the operating system, assign suitable memory and virtual disk space, and attach the downloaded ISO as the virtual optical disc. Start the virtual machine to open the Linux installer. Virtual-machine settings vary by version, so follow the current official instructions.

Expected result

You have either a bootable USB that can start the Linux installer or a virtual machine that boots from the Linux ISO. Do not begin installing Linux onto the real computer unless you understand which disk will be changed.

Practical 3: Change Wallpaper and Language on macOS

Aim

To change the desktop wallpaper and add or select Hindi or Gujarati as a language on a Mac. The exact settings names depend on the macOS version.

Steps to change wallpaper

1. Sign in to the Mac and open the Apple menu in the top-left corner.
2. Open **System Settings** on newer macOS versions. On older versions, open **System Preferences**.
3. Find **Wallpaper** (or Desktop & Screen Saver in older versions).
4. Select one of the built-in wallpapers or choose a picture from a folder.
5. Close Settings and check that the desktop background has changed.

Steps to add Hindi or Gujarati

1. Open the Apple menu and go to **System Settings** → **General** → **Language & Region**. On older macOS versions, Language & Region is inside System Preferences.
2. Find Preferred Languages and press the add (+) button if it is available.
3. Select **Hindi** or **Gujarati**, then choose Add.
4. If macOS asks whether to use the language as the primary language, choose the option that matches the assignment. Changing the primary language may change menus and app language, and macOS may ask you to sign out or restart.
5. If you only need to type in Hindi or Gujarati, add the relevant keyboard/input source in Keyboard settings. Adding a language to the list and changing the entire interface language are not always the same thing.
6. Check the language list and take a screenshot of the changed wallpaper and language setting if your teacher requires proof.

If you do not own a Mac

Do not assume that a Windows computer can run macOS in a normal virtual machine. Apple's macOS licensing and hardware requirements apply. If your school does not provide a Mac, ask whether you may demonstrate this task using a real Mac in the lab or submit a note explaining that you do not have access to one.

Expected result

> The Mac desktop wallpaper is changed and Hindi or Gujarati has been added or selected in Language & Region. If the task is performed on a lab Mac, record the macOS version and what you changed.

Practical 4: Install Ubuntu and Create the User 'gamer'

Aim

To install Ubuntu in a real computer or virtual machine, create a user named **gamer**, enable automatic login where supported, and capture evidence of the setup.

Recommended approach

> For practice, a virtual machine is safer because it avoids changing the main computer's partitions. If using a real computer, back up important files and confirm the correct target disk before installation.

Steps

1. Start the computer from the Ubuntu bootable USB, or start the virtual machine with the Ubuntu ISO attached.
2. Choose **Try or Install Ubuntu** or the equivalent installer option. Follow the on-screen language and keyboard prompts.
3. Connect to Wi-Fi if available and choose the normal installation options required for your setup.
4. At the disk/installation type screen, be very careful. In a virtual machine, select the virtual disk created for that machine. On a real computer, an erase-and-install option can delete existing files and operating systems. Do not proceed unless you have backed up data and understand the selected disk.
5. When asked for account details, enter the name **gamer** as the username. Create a strong password if the installer

asks for one, and keep it private.
6. Continue with the installation. Wait for it to finish, then restart when prompted. If using a USB, remove it when the installer instructs you to do so.
7. Sign in as gamer to confirm the account works. If you created the account with a different display name, check that the actual username is exactly gamer.

Enable automatic login

The setting may differ between Ubuntu versions and desktop environments. On many Ubuntu GNOME installations, open **Settings**, find **System** or **Users**, unlock the settings with an administrator password if required, select the gamer account, and enable **Automatic Login** if the option is available. On some versions, automatic login is configured during installation or through a different user-management screen. If using a lab computer, do not change this security setting without permission.

Security note

Automatic login skips the usual password prompt for the selected account. It may be acceptable for a classroom virtual machine, but it is less secure on a shared or portable real computer. Use it only if the assignment requires it, and turn it off again when no longer needed.

Screenshot evidence

After setup, capture the relevant Users/Automatic Login setting or the login screen if it clearly shows the account. Depending on the Ubuntu configuration, an automatic-login system may not display a normal login screen at startup. If that happens, take a screenshot of the Users settings showing gamer and automatic login enabled, and explain this in your submission. Do not include your password in a screenshot.

Expected result

Ubuntu is installed in the selected real or virtual machine, the account username is gamer, and automatic login is enabled if supported and permitted.

Practical 5: Step-by-Step Guide to Install Windows 10

Aim

To learn the main steps for installing Windows 10, understand disk partitioning, and apply basic privacy and update settings after installation.

Important note about Windows 10

> Microsoft ended standard support for Windows 10 on October 14, 2025. A Windows 10 installation may no longer receive ordinary free security updates unless the device is covered by an applicable Extended Security Updates option or another supported arrangement. For a computer used online, consider a supported operating system, such as a compatible Windows 11 installation, if the hardware allows it. This guide is for learning or a specific assignment requirement.

Before installation

- Back up documents, photos, browser bookmarks, and other important files to a separate drive or trusted cloud storage.
- Check the laptop manufacturer's support page for Wi-Fi, audio, graphics, and storage drivers.
- Keep the laptop plugged into power.
- Use official Microsoft installation media: microsoft.com/software-download/windows10 where available.
- Make sure you have a valid activation method and know whether the device uses a digital licence.
- Creating installation media erases the USB drive. Use a spare drive and check it carefully.

Installation steps

1. Create installation media. On a working Windows computer, open Microsoft's official Windows 10 download page and follow the instructions for the Media Creation Tool or ISO. Use a blank USB drive with enough capacity as specified by Microsoft. Let the tool prepare the installation USB.

2. Boot from the USB. Insert the USB into the target laptop. Restart it and open the boot menu using the key shown by the manufacturer (often a function key or Esc; it varies). Choose the USB installer. If the menu is unfamiliar, check the laptop manufacturer's instructions rather than guessing.

3. Choose setup options. Select the language, time/currency format, and keyboard layout, then select Install now.

4. Enter the product key if requested. Enter a valid key if you have one. If the device has a digital licence, the installer may offer an option to continue and activate later. Select the correct Windows edition that matches the licence.

5. Choose installation type. For a clean installation, the installer usually offers **Custom: Install Windows only**. This is different from an upgrade that attempts to keep apps and files. A clean installation can remove existing files and programs.

6. Understand partitions before changing them. A partition is a section of a storage drive. Windows commonly uses a main Windows partition plus small system/recovery partitions. If you want to keep files or another operating system, stop and get help before deleting or formatting partitions. Deleting partitions can permanently erase data. For a learning virtual machine with no needed data, you can use the virtual disk created for that machine and follow the installer's recommended layout. On a real computer, do not delete partitions merely because their names are unfamiliar.

7. Select the destination. Select the intended drive/partition only after confirming it is the correct disk and you have a verified backup. If the computer has multiple drives, double-check drive sizes. Proceed with installation and allow the computer to restart when required. Do not interrupt power.

8. Complete first-time setup. Follow the region, keyboard, network, account, and privacy prompts. Use a strong password or supported sign-in method. Do not share passwords or recovery keys in screenshots.

9. Install updates and drivers. Once at the desktop, connect to a trusted network. Open **Settings** → **Update & Security** → **Windows Update** and check for updates. Install available updates and restart if requested. Use the laptop maker's official support page for any missing device drivers.

10. Review privacy settings. Open Settings → Privacy and review permissions such as location, camera, microphone, contacts, and diagnostics. Turn off access that you do not need. Some settings are required by apps or managed by an organisation, so read each option before changing it.

11. Check activation and device function. Open Settings → Update & Security → Activation. Test Wi-Fi, sound, brightness keys, touchpad, webcam, and other important hardware. Restore your personal files from the backup only after confirming that the system is stable.

Partitioning tips in simple language

• **Back up first:** partition changes can erase data.
• **Identify the correct disk:** use its size and the installer's information; do not rely on a guessed drive number.
• **Do not delete recovery or system partitions casually:** they may help Windows start or recover.
• **For practice, use a virtual machine:** its virtual disk is separate from the computer's main disk when configured correctly.
• **Stop if unsure:** ask a teacher or technician before formatting or deleting any partition.

Privacy and updates checklist

■ Run Windows Update and restart if needed.
■ Check activation status.
■ Review camera, microphone, location, and app permissions.
■ Use a standard account for everyday tasks when practical; keep administrator access protected.
■ Enable a screen lock and use a strong sign-in method.
■ Use built-in security protection and avoid unknown downloads.
■ Remember Windows 10's support status and plan for a supported OS if this is a daily-use computer.

Reflection: unclear or missing steps

After following the guide, write down any step that was confusing on your particular laptop. Examples include not knowing the boot-menu key, being unsure which partition to select, not finding a driver, or not understanding a privacy option. These details vary by computer model and Windows version, so the best next step is to check the manufacturer's official documentation or ask a teacher before making a risky change.

Observation table

Task	Evidence / result to record	Notes
Windows System Information	Screenshot shows OS, CPU, and RAM	_____
Linux ISO and boot media	ISO downloaded; USB or VM boots installer	_____
macOS settings	Wallpaper changed; Hindi/Gujarati added or selected	_____
Ubuntu gamer account	Username confirmed; auto-login setting checked	_____
Windows 10 guide	Record completed steps and unclear points	_____

Conclusion

This assignment covers basic tasks across Windows, Linux, and macOS. System Information helps identify a computer's hardware and OS. Linux ISO files can be written to a bootable USB or opened in a virtual machine. macOS settings allow users to change wallpaper and language preferences. Ubuntu installation teaches account setup and login configuration. Finally, a careful Windows installation requires backups, correct installation media, safe partition choices, updates, and privacy checks.

The most important rule is to protect existing data. Use official downloads, check the target device and disk carefully, and ask for help whenever an installation or partition screen is unclear.

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